

Part Optimization Log: The Profile page

1. The Baseline Disaster (Before Copilot)

Target: It is hard to read the items in the navigation menu, and the current page that the user is visiting is not highlighted, and the cursor does not change to a pointer while hovering over the items in the navigation menu.

Lighthouse Metrics (CPU 4x Throttling, Network Slow 4G):

Performance: 92.3

Accessibility: 29

First Contentful Paint (FCP): 1.22 s

Largest Contentful Paint (LCP): 1.39 s

Total Blocking Time (TBT): 0

Cumulative Layout Shift: 0

Speed Index: 1.23 s

2. Iterations

Prompt 1: "Make the items in the menu more readable and darker text and more beautiful with Tailwind and keep it on the right side of the page

Highlight the page that the user is visiting

Change the cursor to a pointer when hovering over an item in "

Copilot's Raw Output:

```
1 // App.js
2 <nav className="bg-white border-b border-slate-300 px-5 py-4 flex items-center">
3   <h1 className="text-2xl font-extrabold tracking-tight text-slate-900">My Ecommerce</h1>
4
5   <div className="ml-auto flex gap-2 flex-wrap justify-end items-center">
6     <NavLink
7       tabIndex={7}
8       to="/"
9     >
10      className={({ isActive }) =>
11        `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
12         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
13           ↪ hover:bg-slate-100'}`
14      >
15      Home
16    </NavLink>
17
18    <NavLink
19      tabIndex={8}
20      to="/sales"
21    >
22      className={({ isActive }) =>
23        `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
24         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
25           ↪ hover:bg-slate-100'}`
26      >
27      Sales
28    </NavLink>
29
30    <NavLink
31      tabIndex={2}
32      to="/cart"
33    >
34      className={({ isActive }) =>
35        `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
36         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
37           ↪ hover:bg-slate-100'}`
38      >
39      Cart
40    </NavLink>
```

```

41     tabIndex={10}
42     to="/contact"
43     className={({ isActive }) =>
44       `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
45         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
46         ↪ hover:bg-slate-100'}`
47   }
48   >
49     Contact
50   </NavLink>
51 {isLoggedIn ? (
52   <NavLink
53     tabIndex={11}
54     to="/profile"
55     className={({ isActive }) =>
56       `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
57         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
58         ↪ hover:bg-slate-100'}`
59   >
60     Profile
61   </NavLink>
62 ) : (
63   <>
64     <NavLink
65       tabIndex={1}
66       to="/login"
67       className={({ isActive }) =>
68         `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
69         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
70         ↪ hover:bg-slate-100'}`
71     >
72       Login
73     </NavLink>
74   <NavLink
75     tabIndex={9}
76     to="/register"
77     className={({ isActive }) =>
78       `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
79       ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
80       ↪ hover:bg-slate-100'}`
81     >
82       Register
83     </NavLink>
84   </>
85 )}
86 </div>
87 </nav>

```

Engineering Critique 01: I got an error related to the import of packages and then fixed it with the next prompt.

Prompt 2: "I got an incorrect import. Import the code right"

Copilot's Raw Output:

```
1 // App.js
2 // Use these exact two lines at the top:
3
4 import { useContext } from 'react';
5 import { NavLink, Navigate, Route, Routes, useNavigate } from 'react-router-dom';
```

Engineering Critique 01: ok.

3. The Final Verdict

Lighthouse Metrics (CPU 4x Throttling, Network Slow 4G):
Performance: 93.3 (*Up from 92.3*)
Accessibility: 100 (*Up from 29*)
First Contentful Paint (FCP): 1.26 s (*Up from 1.22*)
Largest Contentful Paint (LCP): 1.26 s (*Down from 1.39*)
Total Blocking Time (TBT): 0
Cumulative Layout Shift: 0
Speed Index: 1.25 s (*Up from 1.23*)

4. Engineering Conclusion

Copilot successfully fixed the navigation menu to a more readable one.

Part Optimization Log: The Home Page

1. The Baseline Disaster (Before Copilot)

Target: 2000 products load in the same time and have high-quality pictures, which makes the loading time very long. The buttons and the texts are not readable, and the pictures do not fit the box in the right dimensions. The page has a gray background, and the search bar, categories, and the menu does not look nice and are hard to read. There is a big banner at the top of the page, which makes the loading time even longer.

Lighthouse Metrics (CPU 4x Throttling, Network Slow 4G):
Performance: 43.6
Accessibility: 29
First Contentful Paint (FCP): 2.1 s
Largest Contentful Paint (LCP): 3.8 s
Total Blocking Time (TBT): 627 ms
Cumulative Layout Shift: 0
Speed Index: 2.76 s

2. Iterations

Prompt 1: "The main div in the project has a colored background and a width of 1024px, and I want to change it to a maximum of 1500px and make the white website background instead of gray. The project uses Tailwind."

Copilot's Raw Output:

```
1 // App.js
2 // replace this line
3 <div className="mx-auto max-w-[1024px] min-h-screen bg-slate-100" aria-hidden="true">
4
5 // with this
6 <div className="mx-auto w-full max-w-[1500px] min-h-screen bg-white" aria-hidden="true">
```

```
1 // Home.js
2 // replace this line
3 <section className="bg-slate-100 min-h-screen text-slate-950">
4
5 // with this
6 <section className="bg-white min-h-screen text-slate-950">
```

Engineering Critique 01: ok

Prompt 2: "Do I need add the viewport meta data in the way it is or another way?
Check if there is any other meta data that needs to be there ?
Update the meta description with the "This is a React shop" and put the lang=en"

Copilot's Raw Output:

```
1 // index.html
2 <!DOCTYPE html>
3 <html lang="en">
4   <head>
5     <meta charset="utf-8" />
6     <link rel="icon" href="%PUBLIC_URL%/favicon.ico" />
7
8     <meta name="viewport" content="width=device-width, initial-scale=1" />
9     <meta name="theme-color" content="#000000" />
10    <meta name="description" content="This is a React shop" />
11
12    <link rel="apple-touch-icon" href="%PUBLIC_URL%/logo192.png" />
13    <link rel="manifest" href="%PUBLIC_URL%/manifest.json" />
14    <script src="https://cdn.tailwindcss.com"></script>
15  </head>
```

Engineering Critique 01: ok

Prompt 3: "Make the grid template column to 4 in the main page
Make the view button and add to cart button more visible with Tailwind classes, and the "add to cart" button should have a blue color background, the view button a light black background, and nicer with Tailwind
The text colors should be darker to make it easier to read
Change the cursor to a pointer when hovering over the view and add to cart buttons
Make the text for the buttons and the link to normal size "

Copilot's Raw Output:

```
1 // H0me.js
2 <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
```

```
1 // H0me.js
2 function ProductCard({ product, onAddToCart, imageVersion }) {
3   const hasDiscount = Number(product.discountPercentage) > 0;
4   const basePrice = Number(product.price) || 0;
5   const discountedPrice = hasDiscount ? getDisplayPrice(product) : null;
6
7   return (
8     <article className="bg-white border border-slate-300 rounded-xl overflow-hidden flex flex-col
9       ↳ shadow-sm hover:shadow-lg transition-shadow">
10       <img
11         src={` ${product.image}?v=${imageVersion}`}
12         alt={product.title}
13         className="h-52 w-full object-contain p-4 bg-slate-50"
14       />
15       <div className="px-4 pb-4 flex flex-col grow">
16         <h3 className="text-lg font-semibold text-slate-900 truncate">{product.title}</h3>
17         <p className="text-base text-slate-700 line-clamp-2 min-h-[48px]">{product.description}</p>
18
19         {hasDiscount ? (
20           <div className="mt-3 mb-4">
21             <p className="text-lg font-medium text-slate-700 line-through">${basePrice.toFixed(2)}</p>
22             <p className="text-xl font-bold text-red-600">${discountedPrice.toFixed(2)}</p>
23             <p className="text-sm font-semibold tracking-wide text-red-600">
24               {product.discountPercentage}% OFF
25             </p>
26           </div>
27         ) : (
28           <p className="text-xl font-bold text-slate-900 mt-3 mb-4">${basePrice.toFixed(2)}</p>
```

```

28     })
29
30     <div className="mt-auto flex gap-2">
31       <Link
32         to={` /product/${product.id}`}
33         className="flex-1 px-4 py-2.5 text-base font-semibold bg-slate-700 text-white text-center
34           ↳ rounded-md hover:bg-slate-800 transition-colors cursor-pointer"
35       >
36         View
37       </Link>
38       <button
39         onClick={() => onAddToCart(product)}
40         className="flex-1 px-4 py-2.5 text-base font-semibold bg-blue-600 text-white rounded-md
41           ↳ hover:bg-blue-700 transition-colors cursor-pointer"
42       >
43         Add to Cart
44       </button>
45     </div>
46   </div>
47 </article>
48 );
49 }

```

```

1 // index.css
2 button {
3   border: 0;
4   background: transparent;
5 }
6
7 a {
8   text-decoration: none;
9 }

```

Engineering Critique 01: This prompt fixed the styling issue in the category area as well.

Prompt 4: "Make the buttons for view and add to cart with less side margin and side padding and a little smaller text."

Copilot's Raw Output:

```

1 // Home.js
2 <div className="mt-auto flex gap-1">
3   <Link
4     to={` /product/${product.id}`}
5     className="flex-1 px-2 py-1.5 text-sm font-medium bg-slate-700 text-white text-center rounded-md
6       ↳ hover:bg-slate-800 transition-colors cursor-pointer"
7   >
8     View
9   </Link>
10  <button
11    onClick={() => onAddToCart(product)}
12    className="flex-1 px-2 py-1.5 text-sm font-medium bg-blue-600 text-white rounded-md hover:bg-blue-700
13      ↳ transition-colors cursor-pointer"
14  >
15    Add to Cart
16  </button>
17 </div>

```

Engineering Critique 01: It change the top and bottom padding, which is not what I wrote, but it made it nicer, so I kept it.

Prompt 5: "Put the original and the discounted price in the same line."

Copilot's Raw Output:

```

1 // Home.js
2 {hasDiscount ? (
3   <div className="mt-3 mb-4">
4     <div className="flex items-baseline gap-2">
5       <p className="text-lg font-medium text-slate-700 line-through">
6         ${basePrice.toFixed(2)}
7       </p>
8       <p className="text-xl font-bold text-red-600">
9         ${discountedPrice.toFixed(2)}
10      </p>
11    </div>
12    <p className="text-sm font-semibold tracking-wide text-red-600">
13      {product.discountPercentage}% OFF
14    </p>
15  </div>
16 ) : (
17   <p className="text-xl font-bold text-slate-900 mt-3 mb-4">${basePrice.toFixed(2)}</p>
18 )}

```

Engineering Critique 01: ok

Prompt 6: "Add Tailwind classes to the search bar with more margin and make the text type to search type
Make the order dropdown nicer with Tailwind classes for dropdown."

Copilot's Raw Output:

```

1 // Home.js
2 <input
3   type="search"
4   value={appState.searchQuery}
5   onChange={(event) => setSearchQuery(event.target.value)}
6   placeholder="Type to search products..."
7   className="w-full sm:w-[300px] mt-2 sm:mt-0 mr-0 sm:mr-4 border border-slate-300 bg-white text-slate-900
8   ↪ placeholder:text-slate-500 rounded-lg px-4 py-2.5 text-sm shadow-sm transition focus:outline-none
9   ↪ focus:ring-2 focus:ring-blue-500 focus:border-blue-500"
10 >

```

```

1 // Home.js
2 <div className="relative min-w-[220px]">
3   <select
4     value={sortOption}
5     onChange={(event) => setSortOption(event.target.value)}
6     className="w-full appearance-none border border-slate-300 bg-white text-slate-900 rounded-lg px-4
7     ↪ py-2.5 pr-10 text-sm font-medium shadow-sm transition hover:border-slate-400 focus:outline-none
8     ↪ focus:ring-2 focus:ring-blue-500 focus:border-blue-500 cursor-pointer"
9     aria-label="Sort products"
10   >
11     <option value="default">Recommended</option>
12     <option value="price-asc">Price: Low to High</option>
13     <option value="price-desc">Price: High to Low</option>
14     <option value="title-asc">Alphabetical (A-Z)</option>
15   </select>
16   <span className="pointer-events-none absolute inset-y-0 right-3 flex items-center text-slate-500">
17     </span>
18 </div>

```

Engineering Critique 01: ok

Prompt 7: "Accept query params: q, category, sort, page, limit Default page=1, limit=24
Apply filtering and sorting on the server
Return: products: [...], page, limit, total, hasMore
Keep existing behavior backward-compatible when no query params are provided."

Make sure sort options include default, price-asc, price-desc, title-asc.
Keep comments normalization logic."

Copilot's Raw Output:

```
1 // server.js
2 if (req.url.startsWith('/api/products') && req.method === 'GET') {
3   const requestUrl = new URL(req.url, `http://localhost:${PORT}`);
4   const { searchParams } = requestUrl;
5
6   const data = await readData();
7   const rawProducts = Array.isArray(data.products) ? data.products : [];
8   const normalizedProducts = rawProducts.map(normalizeProduct);
9
10  const hasMissingCommentsSection = rawProducts.some(
11    (product) => !Array.isArray(product.comments)
12  );
13
14  if (hasMissingCommentsSection) {
15    await writeData({
16      ...data,
17      products: normalizedProducts
18    });
19  }
20
21  const hasQueryParams = ['q', 'category', 'sort', 'page', 'limit'].some((key) =>
22    searchParams.has(key)
23  );
24
25  // Backward-compatible response when no query params are provided
26  if (!hasQueryParams) {
27    respond(res, 200, { products: normalizedProducts });
28    return;
29  }
30
31  const q = (searchParams.get('q') || '').trim().toLowerCase();
32  const category = (searchParams.get('category') || '').trim().toLowerCase();
33  const sort = (searchParams.get('sort') || 'default').trim();
34
35  const pageParam = Number.parseInt(searchParams.get('page') || '1', 10);
36  const limitParam = Number.parseInt(searchParams.get('limit') || '24', 10);
37
38  const page = Number.isFinite(pageParam) && pageParam > 0 ? pageParam : 1;
39  const limit = Number.isFinite(limitParam) && limitParam > 0 ? limitParam : 24;
40
41  let filteredProducts = normalizedProducts;
42
43  if (q) {
44    filteredProducts = filteredProducts.filter((product) => {
45      const haystack = `${product.title} ${product.description}`.toLowerCase();
46      return haystack.includes(q);
47    });
48  }
49
50  if (category && category !== 'all') {
51    filteredProducts = filteredProducts.filter(
52      (product) => (product.category || '').toLowerCase() === category
53    );
54  }
55
56  if (sort === 'price-asc') {
57    filteredProducts = [...filteredProducts].sort(
58      (a, b) => (Number(a.price) || 0) - (Number(b.price) || 0)
59    );
60  } else if (sort === 'price-desc') {
61    filteredProducts = [...filteredProducts].sort(
62      (a, b) => (Number(b.price) || 0) - (Number(a.price) || 0)
63    );
64  } else if (sort === 'title-asc') {
65    filteredProducts = [...filteredProducts].sort((a, b) =>
66      String(a.title || '').localeCompare(String(b.title || ''))
67    );
68  }
```

```

68 }
69 // "default" keeps original order
70
71 const total = filteredProducts.length;
72 const startIndex = (page - 1) * limit;
73 const pagedProducts = filteredProducts.slice(startIndex, startIndex + limit);
74 const hasMore = startIndex + limit < total;
75
76 respond(res, 200, {
77   products: pagedProducts,
78   page,
79   limit,
80   total,
81   hasMore
82 });
83 return;
84 }

```

Engineering Critique 01: ok

Prompt 8: "Add state: productsLoading, productsError, productsPage, productsHasMore, productsTotal

Add method fetchProductsPage(page, append, q, category, sort, limit)

When append=true, append new products instead of replacing

When filters/sort/search change, reset page to 1 and replace list

Expose loadMoreProducts() and resetAndFetchProducts()

Prevent duplicate requests with an in-flight guard

Keep current auth/cart behavior unchanged."

Copilot's Raw Output:

```

1 // AppContext.js
2 import { createContext, useCallback, useEffect, useRef, useState } from 'react';
3 import seedData from './data.json';
4
5 export const AppContext = createContext(null);
6
7 const AUTH_SESSION_KEY = 'authSession';
8 const THIRTY_DAYS_MS = 30 * 24 * 60 * 60 * 1000;
9 const API_BASE_URL = 'http://localhost:4000';
10 const DEFAULT_PRODUCTS_LIMIT = 24;
11
12 const normalizeProduct = (product) => ({
13   ...product,
14   comments: Array.isArray(product.comments) ? product.comments : []
15 });
16
17 const postAuth = async (endpoint, payload) => {
18   let response;
19
20   try {
21     response = await fetch(API_BASE_URL + endpoint, {
22       method: 'POST',
23       headers: {
24         'Content-Type': 'application/json'
25       },
26       body: JSON.stringify(payload)
27     });
28   } catch (error) {
29     const networkError = new Error('HTTP 0');
30     networkError.statusCode = 0;
31     throw networkError;
32   }
33
34   let data = {};
35
36   try {
37     data = await response.json();
38   } catch {

```



```

39   data = {};
40 }
41
42 if (!response.ok) {
43   const httpError = new Error('HTTP ' + response.status);
44   httpError.statusCode = response.status;
45   throw httpError;
46 }
47
48 return data;
49 };
50
51 const saveAuthSession = (email) => {
52   if (typeof window === 'undefined') {
53     return;
54   }
55
56   const payload = {
57     userEmail: email,
58     expiresAt: Date.now() + THIRTY_DAYS_MS
59   };
60
61   window.localStorage.setItem(AUTH_SESSION_KEY, JSON.stringify(payload));
62 };
63
64 const clearAuthSession = () => {
65   if (typeof window === 'undefined') {
66     return;
67   }
68
69   window.localStorage.removeItem(AUTH_SESSION_KEY);
70 };
71
72 const getInitialAuthStatus = () => {
73   if (typeof window === 'undefined') {
74     return {
75       isAuthenticated: false,
76       userEmail: ''
77     };
78   }
79
80   const raw = window.localStorage.getItem(AUTH_SESSION_KEY);
81
82   if (!raw) {
83     return {
84       isAuthenticated: false,
85       userEmail: ''
86     };
87   }
88
89   try {
90     const parsed = JSON.parse(raw);
91
92     if (parsed?.expiresAt && parsed.expiresAt > Date.now() && parsed.userEmail) {
93       return {
94         isAuthenticated: true,
95         userEmail: parsed.userEmail
96       };
97     }
98   } catch {
99     return {
100       isAuthenticated: false,
101       userEmail: ''
102     };
103   }
104
105   clearAuthSession();
106   return {
107     isAuthenticated: false,
108     userEmail: ''
109   };
110 };
111

```

```

112 const getInitialUsers = () => {
113   const defaultUsers = seedData.users || [];
114
115   if (typeof window === 'undefined') {
116     return defaultUsers;
117   }
118
119   const savedUsers = window.localStorage.getItem('registeredUsers');
120
121   if (!savedUsers) {
122     return defaultUsers;
123   }
124
125   try {
126     const parsedUsers = JSON.parse(savedUsers);
127     if (Array.isArray(parsedUsers)) {
128       return parsedUsers;
129     }
130   } catch {
131     return defaultUsers;
132   }
133
134   return defaultUsers;
135 };
136
137 const initialState = {
138   products: (seedData.products || []).map(normalizeProduct),
139   users: getInitialUsers(),
140   userAuthStatus: getInitialAuthStatus(),
141   shoppingCart: [],
142   searchQuery: '',
143   productsLoading: false,
144   productsError: '',
145   productsPage: 1,
146   productsHasMore: true,
147   productsTotal: 0,
148   checkoutFormData: {
149     fullName: '',
150     email: '',
151     password: '',
152     address: '',
153     city: '',
154     zipCode: ''
155   }
156 };
157
158 export function AppProvider({ children }) {
159   const [appState, setAppState] = useState(initialState);
160   const appStateRef = useRef(initialState);
161   const productsRequestInFlightRef = useRef(false);
162
163   useEffect(() => {
164     appStateRef.current = appState;
165   }, [appState]);
166
167   const getEmptyCheckoutForm = () => ({
168     fullName: '',
169     email: '',
170     password: '',
171     address: '',
172     city: '',
173     zipCode: ''
174   });
175
176   const setSearchQuery = (query) => {
177     setAppState((prev) => ({
178       ...prev,
179       searchQuery: query
180     }));
181   };
182
183   const setAuthStatus = (isAuthenticated, userEmail) => {
184     if (isAuthenticated) {

```

```

185     saveAuthSession(userEmail);
186   } else {
187     clearAuthSession();
188   }
189
190   setAppState((prev) => ({
191     ...prev,
192     userAuthStatus: {
193       isAuthenticated,
194       userEmail
195     }
196   }));
197 };
198
199 const fetchProductsPage = useCallback(
200   async ({
201     page = 1,
202     append = false,
203     q = '',
204     category = 'All',
205     sort = 'default',
206     limit = DEFAULT_PRODUCTS_LIMIT
207   } = {}) => {
208     if (productsRequestInFlightRef.current) {
209       return null;
210     }
211
212     productsRequestInFlightRef.current = true;
213
214     setAppState((prev) => ({
215       ...prev,
216       productsLoading: true,
217       productsError: ''
218     }));
219
220     const safePage = Number.isFinite(Number(page)) && Number(page) > 0 ? Number(page) : 1;
221     const safeLimit = Number.isFinite(Number(limit)) && Number(limit) > 0 ? Number(limit) :
222       ↪ DEFAULT_PRODUCTS_LIMIT;
223
224     const params = new URLSearchParams();
225     params.set('page', String(safePage));
226     params.set('limit', String(safeLimit));
227     params.set('sort', sort || 'default');
228
229     const normalizedQ = (q || '').trim();
230     const normalizedCategory = (category || '').trim();
231
232     if (normalizedQ) {
233       params.set('q', normalizedQ);
234     }
235
236     if (normalizedCategory && normalizedCategory.toLowerCase() !== 'all') {
237       params.set('category', normalizedCategory);
238     }
239
240     try {
241       const response = await fetch(API_BASE_URL + '/api/products?' + params.toString());
242       if (!response.ok) {
243         throw new Error('HTTP ' + response.status);
244       }
245
246       const payload = await response.json();
247       const nextProducts = Array.isArray(payload.products)
248         ? payload.products.map(normalizeProduct)
249         : [];
250
251       const total = Number(payload.total) || 0;
252       const hasMore =
253         typeof payload.hasMore === 'boolean'
254           ? payload.hasMore
255           : safePage * safeLimit < total;
256
257       setAppState((prev) => ({

```

```

257     ...prev,
258     products: append ? [...prev.products, ...nextProducts] : nextProducts,
259     productsLoading: false,
260     productsError: '',
261     productsPage: Number(payload.page) || safePage,
262     productsHasMore: hasMore,
263     productsTotal: total
264   }));
265
266   return payload;
267 } catch (error) {
268   setAppState((prev) => ({
269     ...prev,
270     productsLoading: false,
271     productsError: 'Failed to load products'
272   }));
273   return null;
274 } finally {
275   productsRequestInFlightRef.current = false;
276 }
277 },
278 []
279 );
280
281 const resetAndFetchProducts = useCallback(
282   async ({
283     q = '',
284     category = 'All',
285     sort = 'default',
286     limit = DEFAULT_PRODUCTS_LIMIT
287   } = {}) => {
288     setAppState((prev) => ({
289       ...prev,
290       searchQuery: q,
291       productsPage: 1,
292       productsHasMore: true
293     }));
294
295     return fetchProductsPage({
296       page: 1,
297       append: false,
298       q,
299       category,
300       sort,
301       limit
302     });
303   },
304   [fetchProductsPage]
305 );
306
307 const loadMoreProducts = useCallback(
308   async ({
309     q = appStateRef.current.searchQuery,
310     category = 'All',
311     sort = 'default',
312     limit = DEFAULT_PRODUCTS_LIMIT
313   } = {}) => {
314     const snapshot = appStateRef.current;
315
316     if (snapshot.productsLoading || !snapshot.productsHasMore) {
317       return null;
318     }
319
320     const nextPage = (snapshot.productsPage || 1) + 1;
321
322     return fetchProductsPage({
323       page: nextPage,
324       append: true,
325       q,
326       category,
327       sort,
328       limit
329     });

```

```

330     },
331     [fetchProductsPage]
332 );
333
334 const loginUser = async (email, password) => {
335     const payload = await postAuth('/api/login', { email, password });
336
337     setAuthStatus(true, payload.user.email);
338     return payload.user;
339 };
340
341 const registerUser = async (name, email, password) => {
342     const payload = await postAuth('/api/register', { name, email, password });
343
344     setAppState((prev) => ({
345         ...prev,
346         users: payload.users || prev.users
347     }));
348
349     setAuthStatus(true, payload.user.email);
350     return payload.user;
351 };
352
353 const placeOrder = async (orderPayload) => {
354     const payload = await postAuth('/api/place-order', orderPayload);
355
356     setAppState((prev) => ({
357         ...prev,
358         users: payload.users || prev.users,
359         shoppingCart: [],
360         checkoutFormData: getEmptyCheckoutForm(),
361         userAuthStatus: {
362             isAuthenticated: true,
363             userEmail: payload.user?.email || prev.userAuthStatus.userEmail
364         }
365     }));
366
367     if (payload.user?.email) {
368         saveAuthSession(payload.user.email);
369     }
370
371     return payload;
372 };
373
374 const logoutUser = () => {
375     setAuthStatus(false, '');
376 };
377
378 const addProductComment = async (commentPayload) => {
379     const payload = await postAuth('/api/product-comments', commentPayload);
380
381     setAppState((prev) => ({
382         ...prev,
383         products: Array.isArray(payload.products)
384             ? payload.products.map(normalizeProduct)
385             : prev.products
386     }));
387
388     return payload.comment;
389 };
390
391 const addToCart = (product) => {
392     setAppState((prev) => ({
393         ...prev,
394         shoppingCart: [...prev.shoppingCart, product]
395     }));
396     window.alert('Added to cart');
397 };
398
399 const removeFromCart = (productId) => {
400     setAppState((prev) => ({
401         ...prev,
402         shoppingCart: prev.shoppingCart.filter((item, index) => {

```

```

403     if (item.id !== productId) {
404         return true;
405     }
406
407     return prev.shoppingCart.findIndex((cartItem) => cartItem.id === productId) !== index;
408 })
409 });
410 window.alert('Removed from cart');
411 };
412
413 const updateCheckoutField = (fieldName, value) => {
414     setAppState((prev) => ({
415         ...prev,
416         checkoutFormData: {
417             ...prev.checkoutFormData,
418             [fieldName]: value
419         }
420     }));
421 };
422
423 const resetCheckoutForm = () => {
424     setAppState((prev) => ({
425         ...prev,
426         checkoutFormData: getEmptyCheckoutForm()
427     }));
428 };
429
430 useEffect(() => {
431     let isMounted = true;
432
433     fetch(API_BASE_URL + '/api/users')
434     .then((response) => response.json())
435     .then((payload) => {
436         if (!isMounted) {
437             return;
438         }
439
440         if (Array.isArray(payload.users)) {
441             setAppState((prev) => ({
442                 ...prev,
443                 users: payload.users
444             }));
445         }
446     })
447     .catch(() => {});
448
449     fetchProductsPage({
450         page: 1,
451         append: false,
452         q: '',
453         category: 'All',
454         sort: 'default',
455         limit: DEFAULT_PRODUCTS_LIMIT
456     });
457
458     return () => {
459         isMounted = false;
460     };
461 }, [fetchProductsPage]);
462
463 const contextValue = {
464     appState,
465     setAuthStatus,
466     loginUser,
467     registerUser,
468     placeOrder,
469     logoutUser,
470     setSearchQuery,
471     fetchProductsPage,
472     loadMoreProducts,
473     resetAndFetchProducts,
474     addToCart,
475     removeFromCart,

```

```

476     addProductComment,
477     updateCheckoutField,
478     resetCheckoutForm
479   };
480
481   return <AppContext.Provider value={contextValue}>{children}</AppContext.Provider>;
482 }

```

Engineering Critique 01: ok

Prompt 9: "Remove client-side heavy filtering/sorting over full array

Use context methods to request products by page

Add a sentinel div at the bottom of the grid

Use IntersectionObserver to call loadMoreProducts() when sentinel enters viewport

Show loading spinner/skeleton while fetching next page

Show "No more products" when hasMore is false

Preserve existing category/sort/search controls."

Copilot's Raw Output:

```

1 // Home.js
2 - import { getDisplayPrice, sortProducts } from './utils/productSorting';
3 + import { getDisplayPrice } from './utils/productSorting';

```

```

1 // Home.js
2 - function ProductCard({ product, onAddToCart, imageVersion }) {
3 + function ProductCard({ product, onAddToCart }) {

```

```

1 // Home.js
2 - src={` ${product.image}?v=${imageVersion}`}
3 + src={product.image}
4 + loading="lazy"
5 + decoding="async"

```

```

1 // Home.js
2 - const { appState, setSearchQuery, addToCart } = useContext(AppContext);
3 + const {
4 +   appState,
5 +   setSearchQuery,
6 +   addToCart,
7 +   loadMoreProducts,
8 +   resetAndFetchProducts
9 + } = useContext(AppContext);
10
11 const [selectedCategory, setSelectedCategory] = useState('All');
12 const [sortOption, setSortOption] = useState('default');
13 - const imageVersion = Date.now();
14 + const sentinelRef = useRef(null);
15 + const didInitFiltersRef = useRef(false);

```

```

1 // Home.js
2 - const searchFiltered = appState.products.filter((product) =>
3 -   product.title.toLowerCase().includes(appState.searchQuery.toLowerCase())
4 - );
5 -
6 - const filteredProducts =
7 -   selectedCategory === 'All'
8 -     ? searchFiltered
9 -     : searchFiltered.filter(
10 -       (product) => product.category?.toLowerCase() === selectedCategory.toLowerCase()
11 -     );

```

```

12 -
13 - const sortedProducts = sortProducts(filteredProducts, sortOption);

```

```

1 // Home.js
2 useEffect(() => {
3   if (!didInitFiltersRef.current) {
4     didInitFiltersRef.current = true;
5     return;
6   }
7
8   resetAndFetchProducts({
9     q: appState.searchQuery,
10    category: selectedCategory,
11    sort: sortOption
12  });
13 }, [appState.searchQuery, selectedCategory, sortOption, resetAndFetchProducts]);

```

```

1 // Home.js
2 useEffect(() => {
3   if (!sentinelRef.current) {
4     return;
5   }
6
7   const node = sentinelRef.current;
8   const observer = new IntersectionObserver(
9     (entries) => {
10      const firstEntry = entries[0];
11      if (!firstEntry?.isIntersecting) {
12        return;
13      }
14
15      if (appState.productsLoading || !appState.productsHasMore) {
16        return;
17      }
18
19      loadMoreProducts({
20        q: appState.searchQuery,
21        category: selectedCategory,
22        sort: sortOption
23      });
24    },
25    {
26      root: null,
27      rootMargin: '300px 0px',
28      threshold: 0
29    }
30  );
31
32  observer.observe(node);
33
34  return () => {
35    observer.disconnect();
36  };
37 }, [
38   appState.productsLoading,
39   appState.productsHasMore,
40   appState.searchQuery,
41   selectedCategory,
42   sortOption,
43   loadMoreProducts
44 ]);

```

```

1 // Home.js
2 const showInitialLoading = appState.productsLoading && appState.products.length === 0;
3 const showNextPageLoading = appState.productsLoading && appState.products.length > 0;
4 const showNoProducts = !appState.productsLoading && appState.products.length === 0;
5 const showNoMore = !appState.productsLoading && appState.products.length > 0 && !appState.productsHasMore;

```



```

1 // Home.js
2 - {sortedProducts.length === 0 ? (
3 + {showInitialLoading ? (
4 +   <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
5 +     {Array.from({ length: 8 }).map((_, index) => (
6 +       <div
7 +         key={index}
8 +         className="h-80 rounded-xl border border-slate-200 bg-slate-100 animate-pulse"
9 +       >
10 +     )}}
11 +   </div>
12 + ) : showNoProducts ? (
13 +   <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700 shadow-sm">
14 +     No products found.
15 +   </div>
16 + ) : (
17 -   <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
18 -     {sortedProducts.map((product) => (
19 +     <>
20 +       <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
21 +         {appState.products.map((product) => (
22 +           <ProductCard
23 +             key={product.id}
24 +             product={product}
25 +             onAddToCart={addToCart}
26 +             imageVersion={imageVersion}
27 +           >
28 -         )}}
29 -       </div>
30 +       </div>
31 +     </div>
32 +
33 +     <div ref={sentinelRef} className="h-10" />
34 +
35 +     {showNextPageLoading && (
36 +       <div className="flex justify-center py-4">
37 +         <div className="h-8 w-8 rounded-full border-4 border-slate-300 border-t-blue-600 animate-spin"
38 +       >
39 +     </div>
40 +   )}
41 +   {showNoMore && (
42 +     <p className="text-center text-sm text-slate-500 py-4">
43 +       No more products.
44 +     </p>
45 +   )}
46 + </>
47 )}

```

```

1 // sever.js
2 - if (req.url === '/api/products' && req.method === 'GET') {

```

Engineering Critique 01: I got this error: Line 93:26: 'sortProducts' is not defined no-undef

Prompt 10: "'sortProducts' is not defined no-undef"

Copilot's Raw Output:

```

1 // Home.js
2 const categories = [
3   { label: 'All', value: 'All' },
4   { label: 'Men's Clothing', value: 'men's clothing' },
5   { label: 'Women's Clothing', value: 'women's clothing' },
6   { label: 'Jewelery', value: 'jewelery' },
7   { label: 'Electronics', value: 'electronics' }
8 ];
9 -

```

```

10 - const searchFiltered = appState.products.filter((product) =>
11 -   product.title.toLowerCase().includes(appState.searchQuery.toLowerCase())
12 - );
13 -
14 - const filteredProducts =
15 -   selectedCategory === 'All'
16 -     ? searchFiltered
17 -     : searchFiltered.filter(
18 -       (product) => product.category?.toLowerCase() === selectedCategory.toLowerCase()
19 -     );
20 -
21 - const sortedProducts = sortProducts(filteredProducts, sortOption);

```

Engineering Critique 01: Products loads 2 times

Prompt 11: "I load the home page and it shows all products for some seconds at first and then shows a message "loading more products" and shows the product again."

Copilot's Raw Output:

```

1 // AppContext.js
2 @@
3 const initialState = {
4 - products: (seedData.products || []).map(normalizeProduct),
5 + products: [],
6 @@
7 - productsLoading: false,
8 + productsLoading: true,

```

```

1 // Home.js
2 @@
3   {showInitialLoading ? (
4 -     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
5 -     shadow-sm">
6 -       Loading...
7 -     </div>
8 -   ) : showNextPageLoading ? (
9 -     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
10 -     shadow-sm">
11 -       Loading more products...
12 -     </div>
13 -   ) : showNoProducts ? (
14 -     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
15 -     shadow-sm">
16 -       No products found.
17 -     </div>
18 -   ) : showNoMore ? (
19 -     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
20 -     shadow-sm">
21 -       No more products to load.
22 -     </div>
23 -   ) : (
24 +   {showInitialLoading ? (
25 +     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
26 +     shadow-sm">
27 +       Loading...
28 +     </div>
29 +   ) : showNoProducts ? (
30 +     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
31 +     shadow-sm">
32 +       No products found.
33 +     </div>
34 +   ) : (
35 +     <>
36 +       <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
37 +         {appState.products.map((product) => (
38 +           <ProductCard
39 +             key={product.id}

```

```

34         product={product}
35         onAddToCart={addToCart}
36     />
37     )}}
38 </div>
39 <div ref={sentinelRef} className="h-10" />
40 {showNextPageLoading && (
41     <div className="flex justify-center py-4">
42         <div className="h-8 w-8 rounded-full border-4 border-slate-300 border-t-blue-600
43             ↪ animate-spin" />
44     </div>
45 )}
46 {showNoMore && (
47     <p className="text-center text-sm text-slate-500 py-4">
48         No more products.
49     </p>
50 )}
51 </>
52 )}

```

```

1 // Home.js
2 @@
3 -     rootMargin: '300px 0px',
4 +     rootMargin: '80px 0px',

```

Engineering Critique 01: No products shown in the page

Prompt 12: "No product shows in the home page and just message "No products found"

Copilot's Raw Output:

```

1 // AppContext.js
2 @@
3 -     try {
4 -         const response = await fetch(API_BASE_URL + '/api/products?' + params.toString());
5 -         if (!response.ok) {
6 -             throw new Error('HTTP ' + response.status);
7 -         }
8 -
9 -         const payload = await response.json();
10 +     try {
11 +         let payload;
12 +         const primaryResponse = await fetch(API_BASE_URL + '/api/products?' + params.toString());
13 +
14 +         if (primaryResponse.ok) {
15 +             payload = await primaryResponse.json();
16 +         } else if (safePage === 1 && !append) {
17 +             // Fallback for older backend handlers that only support /api/products
18 +             const fallbackResponse = await fetch(API_BASE_URL + '/api/products');
19 +             if (!fallbackResponse.ok) {
20 +                 throw new Error('HTTP ' + primaryResponse.status);
21 +             }
22 +
23 +             const fallbackPayload = await fallbackResponse.json();
24 +             const fallbackProducts = Array.isArray(fallbackPayload.products) ? fallbackPayload.products :
25 ↪     [];
26 +
27 +             payload = {
28 +                 products: fallbackProducts,
29 +                 page: 1,
30 +                 limit: safeLimit,
31 +                 total: fallbackProducts.length,
32 +                 hasMore: fallbackProducts.length > safeLimit
33 +             };
34 +         } else {
35 +             throw new Error('HTTP ' + primaryResponse.status);
36 +         }

```

```

37     const nextProducts = Array.isArray(payload.products)
38       ? payload.products.map(normalizeProduct)
39       : [];

```

```

1 // Home.js
2 @@
3 -       ) : showNoProducts ? (
4 +       ) : showError ? (
5 +       <div className="rounded-xl border border-red-300 bg-red-50 p-8 text-center text-red-700
6 +       shadow-sm">
7 +       {appState.productsError}
8 +       </div>
9 +       ) : showNoProducts ? (

```

Engineering Critique 01: Categories does not update.

Prompt 13: "Fix the 404 for GET /api/products?page=... in server.js."

Copilot's Raw Output:

```

1 // server.js
2 @@
3 - if (req.url.startsWith('/api/products') && req.method === 'GET') {
4 -   const requestUrl = new URL(req.url, `http://localhost:${PORT}`);
5 + const requestUrl = new URL(req.url, `http://localhost:${PORT}`);
6 + const { pathname, searchParams } = requestUrl;
7 +
8 + if (pathname === '/api/products' && req.method === 'GET') {
9 -   const { searchParams } = requestUrl;

```

Engineering Critique 01: ok

Prompt 14: "Debounce search input by 300ms

On debounced value change, reset page and fetch first page from server

Do not fetch on every keystroke

Keep input responsive and controlled

Cancel stale effects on unmount/change."

Copilot's Raw Output:

```

1 // Home.js
2 @@
3 function Home() {
4 @@
5   const [selectedCategory, setSelectedCategory] = useState('All');
6   const [sortOption, setSortOption] = useState('default');
7 + const [localSearchQuery, setLocalSearchQuery] = useState(appState.searchQuery || '');
8 + const [debouncedSearchQuery, setDebouncedSearchQuery] = useState(appState.searchQuery || '');
9   const sentinelRef = useRef(null);
10  const didInitFiltersRef = useRef(false);

```

```

1 // Home.js
2 @@
3 const categories = [
4 @@
5 ];
6 +
7 + useEffect(() => {
8 +   const timeoutId = window.setTimeout(() => {
9 +     setDebouncedSearchQuery(localSearchQuery);
10 +     setSearchQuery(localSearchQuery);
11 +   }, 300);

```

```

12 +
13 +   return () => {
14 +     window.clearTimeout(timeoutId);
15 +   };
16 + }, [localSearchQuery, setSearchQuery]);

```

```

1 // Home.js
2 @@
3 -   resetAndFetchProducts({
4 -     q: appState.searchQuery,
5 +   resetAndFetchProducts({
6 +     q: debouncedSearchQuery,
7     category: selectedCategory,
8     sort: sortOption
9   });
10 - }, [appState.searchQuery, selectedCategory, sortOption, resetAndFetchProducts]);
11 + }, [debouncedSearchQuery, selectedCategory, sortOption, resetAndFetchProducts]);

```

```

1 // Home.js
2 @@
3 -   loadMoreProducts({
4 -     q: appState.searchQuery,
5 +   loadMoreProducts({
6 +     q: debouncedSearchQuery,
7     category: selectedCategory,
8     sort: sortOption
9   });
10 @@
11 -   appState.searchQuery,
12 +   debouncedSearchQuery,

```

```

1 // Home.js
2 @@
3     <input
4       type="search"
5 -     value={appState.searchQuery}
6 -     onChange={(event) => setSearchQuery(event.target.value)}
7 +     value={localSearchQuery}
8 +     onChange={(event) => setLocalSearchQuery(event.target.value)}
9     placeholder="Type to search products..."
10    className="w-full sm:w-[300px] mt-2 sm:mt-0 mr-0 sm:mr-4 border border-slate-300 bg-white
11      ↪ text-slate-900 placeholder:text-slate-500 rounded-lg px-4 py-2.5 text-sm shadow-sm
12      ↪ transition focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-blue-500"
13  />

```

Engineering Critique 01: ok

Prompt 15: "Wrap ProductCard in React.memo
Remove image URL cache busting with Date.now
Keep existing visual styles and behavior
Ensure Add to Cart still works."

Copilot's Raw Output:

```

1 // Home.js
2 @@
3 -import { useContext, useState, useRef, useEffect } from 'react';
4 +import { memo, useContext, useState, useRef, useEffect } from 'react';
5 @@
6 -function ProductCard({ product, onAddToCart }) {
7 +const ProductCard = memo(function ProductCard({ product, onAddToCart }) {
8 @@
9 -}
10 +});

```

Engineering Critique 01: Forgot to remove extra variable 'showNoProducts.'

Prompt 16: "Remove extra variable 'showNoProducts' is assigned a value but never used."

Copilot's Raw Output:

```
1 // Home.js
2 @@
3 - const showNoProducts = !appState.productsLoading && appState.products.length === 0;
```

Engineering Critique 01: ok

Prompt 17: "Add retry UI when request fails
Add loading skeleton cards while first page loads
Add small "Loading more..." indicator for next pages
Avoid duplicate loads when user scrolls fast."

Copilot's Raw Output:

```
1 // Home.js
2 @@
3 -import { memo, useContext, useState, useRef, useEffect } from 'react';
4 +import { memo, useContext, useState, useRef, useEffect } from 'react';
5 @@
6   const [localSearchQuery, setLocalSearchQuery] = useState(appState.searchQuery || '');
7   const [debouncedSearchQuery, setDebouncedSearchQuery] = useState(appState.searchQuery || '');
8   const sentinelRef = useRef(null);
9 + const observerLoadPendingRef = useRef(false);
10  const didInitFiltersRef = useRef(false);
```

```
1 // Home.js
2 @@
3   useEffect(() => {
4     @@
5     - const observer = new IntersectionObserver(
6     -   (entries) => {
7     + const observer = new IntersectionObserver(
8     +   (entries) => {
9       const firstEntry = entries[0];
10      if (!firstEntry?.isIntersecting) {
11        return;
12      }
13
14      - if (appState.productsLoading || !appState.productsHasMore) {
15      + if (observerLoadPendingRef.current || appState.productsLoading || !appState.productsHasMore) {
16        return;
17      }
18
19      - loadMoreProducts({
20      -   q: debouncedSearchQuery,
21      -   category: selectedCategory,
22      -   sort: sortOption
23      - });
24      + observerLoadPendingRef.current = true;
25      + Promise.resolve(
26      +   loadMoreProducts({
27      +     q: debouncedSearchQuery,
28      +     category: selectedCategory,
29      +     sort: sortOption
30      +   })
31      + ).finally(() => {
32      +   observerLoadPendingRef.current = false;
33      + });
34    },
35    {
```

```

36     root: null,
37     rootMargin: '80px 0px',
38     threshold: 0
39   }, [
40     appState.productsLoading,
41     appState.productsHasMore,
42     appState.searchQuery,
43     debouncedSearchQuery,
44     selectedCategory,
45     sortOption,
46     loadMoreProducts
47   ]);
48

```

```

1  // Home.js
2  @@
3  -   {showInitialLoading ? (
4  -     <div className="rounded-xl border border-slate-300 bg-white p-8 text-center text-slate-700
5  -     shadow-sm">
6  -       Loading...
7  -     </div>
8  +   {showInitialLoading ? (
9  +     <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-5">
10 +       {Array.from({ length: 8 }).map((_, index) => (
11 +         <article
12 +           key={index}
13 +           className="bg-white border border-slate-300 rounded-xl overflow-hidden shadow-sm
14 +           animate-pulse"
15 +         >
16 +           <div className="h-52 w-full bg-slate-100" />
17 +           <div className="px-4 py-4 space-y-3">
18 +             <div className="h-5 w-3/4 bg-slate-200 rounded" />
19 +             <div className="h-4 w-full bg-slate-100 rounded" />
20 +             <div className="h-4 w-2/3 bg-slate-100 rounded" />
21 +             <div className="h-8 w-full bg-slate-200 rounded mt-4" />
22 +           </div>
23 +         </article>
24 +       )]}
25 +     </div>
26 +   ) : showError ? (
27 +     <div className="rounded-xl border border-red-300 bg-red-50 p-8 text-center text-red-700
28 +     shadow-sm">
29 +       {appState.productsError}
30 +       <p>{appState.productsError}</p>
31 +       <button
32 +         className="mt-4 px-4 py-2 text-sm font-medium rounded bg-slate-900 text-white
33 +         hover:bg-slate-800 transition-colors cursor-pointer"
34 +         onClick={() =>
35 +           resetAndFetchProducts({
36 +             q: debouncedSearchQuery,
37 +             category: selectedCategory,
38 +             sort: sortOption
39 +           })
40 +         }
41 +       >
42 +         Retry
43 +       </button>
44 +     </div>
45 +   ) : (
46 +     @@
47 +     {showNextPageLoading && (
48 +       <div className="flex justify-center py-4">
49 +         <div className="flex items-center justify-center gap-2 py-4">
50 +           <div className="h-8 w-8 rounded-full border-4 border-slate-300 border-t-blue-600
51 +           animate-spin" />
52 +           <p className="text-xs text-slate-600">Loading more...</p>
53 +         </div>
54 +       </div>
55 +     )}
56 +   )
57 + }
58 +

```

Engineering Critique 01: ok

Prompt 18: "Keep existing endpoint path /api/products.
 Keep current query params: q, category, sort, page, limit.
 Return only card fields for the list response: id, title, price, discountPercentage,
 category, image, thumbnail, imageSet Do not include comments in list payload.
 Keep pagination shape: products, page, limit, total, hasMore
 Keep backward compatibility for current filters/sort values.
 Keep Product Details behavior unchanged by exposing a separate full product endpoint: GET /api/products/:id
 returns full product with comments and original image.
 Add small helper(s) to derive thumbnail URLs from current image path.
 Preserve existing auth/cart/order routes unchanged."

Copilot's Raw Output:

```

1 // server.js
2 --- a/server.js
3 +++ b/server.js
4 @@
5   const normalizeProduct = (product) => ({
6     ...product,
7     comments: Array.isArray(product.comments) ? product.comments : []
8   });
9 +
10 +const stripImageExtension = (imagePath = '') => {
11 +  const ext = path.extname(imagePath || '');
12 +  if (!ext) {
13 +    return imagePath || '';
14 +  }
15 +  return imagePath.slice(0, -ext.length);
16 +};
17 +
18 +const deriveProductImageFields = (imagePath = '') => {
19 +  const base = stripImageExtension(imagePath || '');
20 +  const ext = path.extname(imagePath || '') || '.jpg';
21 +
22 +  if (!base) {
23 +    return {
24 +      thumbnail: '',
25 +      imageSet: ''
26 +    };
27 +  }
28 +
29 +  const thumbnail = `${base}-thumb${ext}`;
30 +  const imageSet = `${base}-thumb.avif 1x, ${base}-thumb.webp 1x, ${thumbnail} 1x`;
31 +
32 +  return {
33 +    thumbnail,
34 +    imageSet
35 +  };
36 +};
37 +
38 +const toProductCard = (product) => {
39 +  const image = product?.image || '';
40 +  const { thumbnail, imageSet } = deriveProductImageFields(image);
41 +
42 +  return {
43 +    id: product.id,
44 +    title: product.title,
45 +    price: product.price,
46 +    discountPercentage: product.discountPercentage,
47 +    category: product.category,
48 +    image,
49 +    thumbnail,
50 +    imageSet
51 +  };
52 +};
53 @@
54   const requestUrl = new URL(req.url, `http://localhost:${PORT}`);
55   const { pathname, searchParams } = requestUrl;
56 +
57 +  if (pathname.startsWith('/api/products/') && req.method === 'GET') {

```



```

58 +   const productId = Number(pathname.split('/').pop());
59 +
60 +   if (!Number.isFinite(productId) || productId <= 0) {
61 +     respond(res, 400, { message: 'Invalid product ID.' });
62 +     return;
63 +   }
64 +
65 +   const data = await readData();
66 +   const products = (Array.isArray(data.products) ? data.products : []).map(normalizeProduct);
67 +   const product = products.find((entry) => Number(entry.id) === productId);
68 +
69 +   if (!product) {
70 +     respond(res, 404, { message: 'Product not found.' });
71 +     return;
72 +   }
73 +
74 +   respond(res, 200, { product });
75 +   return;
76 + }
77 @@
78 if (pathname === '/api/products' && req.method === 'GET') {
79   const { searchParams } = requestUrl;
80 @@
81   const hasQueryParams = ['q', 'category', 'sort', 'page', 'limit'].some((key) =>
82     searchParams.has(key)
83   );
84
85   // Backward-compatible response when no query params are provided
86   if (!hasQueryParams) {
87     respond(res, 200, { products: normalizedProducts });
88     return;
89   }
90
91   const q = (searchParams.get('q') || '').trim().toLowerCase();
92   const category = (searchParams.get('category') || '').trim().toLowerCase();
93   const sort = (searchParams.get('sort') || 'default').trim();
94 @@
95   const total = filteredProducts.length;
96   const startIndex = (page - 1) * limit;
97   const pagedProducts = filteredProducts.slice(startIndex, startIndex + limit);
98 +   const pagedProducts = filteredProducts
99 +     .slice(startIndex, startIndex + limit)
100 +     .map(toProductCard);
101   const hasMore = startIndex + limit < total;
102
103   respond(res, 200, {
104     products: pagedProducts,
105     page,
106     limit,
107     total,
108     hasMore
109   });
110   return;
111 }

```

Engineering Critique 01: ok

Prompt 19: "Generate thumbnail for picture (350px*210px) and use them in home page
In Home grid, use thumbnail-first sources (avif/webp/jpg fallback) from product. thumbnail or product.imageSet.

Keep product. image (original/full) for Product Details only.

Add width and height attributes (or equivalent stable container dimensions) to reduce layout shift.

Keep loading=lazy and decoding=async on grid images.

Keep existing card styles and add to Cart behavior.

If thumbnail fails, fallback to original image gracefully."

Copilot's Raw Output:

```

1 // Home.js

```

```

2 const splitImageSetByFormat = (imageSet = '') => {
3   const urls = String(imageSet)
4     .split(',')
5     .map((chunk) => chunk.trim().split(/\s+/)[0])
6     .filter(Boolean);
7
8   return {
9     avif: urls.find((url) => url.toLowerCase().endsWith('.avif')) || '',
10    webp: urls.find((url) => url.toLowerCase().endsWith('.webp')) || '',
11    jpg: urls.find((url) => /\.?(jpe?g|png)$/i.test(url)) || ''
12  };
13 };
14
15 const deriveThumbFormatsFromThumbnail = (thumbnail = '') => {
16   if (!thumbnail) {
17     return { avif: '', webp: '', jpg: '' };
18   }
19
20   const base = thumbnail.replace(/\.([~/.]+)$/, '');
21   return {
22     avif: `${base}.avif`,
23     webp: `${base}.webp`,
24     jpg: thumbnail
25   };
26 };
27
28 const getCardImageSources = (product) => {
29   const fromImageSet = splitImageSetByFormat(product?.imageSet || '');
30   const fromThumbnail = deriveThumbFormatsFromThumbnail(product?.thumbnail || '');
31
32   return {
33     avif: fromImageSet.avif || fromThumbnail.avif,
34     webp: fromImageSet.webp || fromThumbnail.webp,
35     jpg: fromImageSet.jpg || fromThumbnail.jpg || product?.thumbnail || product?.image || '',
36     original: product?.image || ''
37   };
38 };

```

```

1 // Home.js
2 const ProductCard = memo(function ProductCard({ product, onAddToCart }) {
3   const hasDiscount = Number(product.discountPercentage) > 0;
4   const basePrice = Number(product.price) || 0;
5   const discountedPrice = hasDiscount ? getDisplayPrice(product) : null;
6
7   const [useOriginalImage, setUseOriginalImage] = useState(false);
8   const sources = useMemo(() => getCardImageSources(product), [product]);
9
10  const imageSrc = useOriginalImage
11    ? (sources.original || sources.jpg)
12    : (sources.jpg || sources.original);
13
14  const handleImageError = () => {
15    if (!useOriginalImage && sources.original) {
16      setUseOriginalImage(true);
17    }
18  };
19
20  return (
21    <article className="bg-white border border-slate-300 rounded-xl overflow-hidden flex flex-col
22      ↳ shadow-sm hover:shadow-lg transition-shadow">
23      <div className="h-52 w-full bg-slate-50">
24        <picture>
25          {!useOriginalImage && sources.avif ? (
26            <source srcSet={sources.avif} type="image/avif" />
27          ) : null}
28          {!useOriginalImage && sources.webp ? (
29            <source srcSet={sources.webp} type="image/webp" />
30          ) : null}
31          <img
32            src={imageSrc}
33            alt={product.title}
34            className="h-52 w-full object-contain p-4"

```

```

34     width={350}
35     height={210}
36     loading="lazy"
37     decoding="async"
38     onError={handleImageError}
39   />
40   </picture>
41 </div>
42
43 <div className="px-4 pb-4 flex flex-col grow">
44   <h3 className="text-lg font-semibold text-slate-900 truncate">{product.title}</h3>
45   <p className="text-base text-slate-700 line-clamp-2 min-h-[48px]">{product.description}</p>
46
47   {hasDiscount ? (
48     <div className="mt-3 mb-4">
49       <div className="flex items-baseline gap-2">
50         <p className="text-lg font-medium text-slate-700 line-through">
51           ${basePrice.toFixed(2)}
52         </p>
53         <p className="text-xl font-bold text-red-600">
54           ${discountedPrice.toFixed(2)}
55         </p>
56       </div>
57       <p className="text-sm font-semibold tracking-wide text-red-600">
58         {product.discountPercentage}% OFF
59       </p>
60     </div>
61   ) : (
62     <p className="text-xl font-bold text-slate-900 mt-3 mb-4">${basePrice.toFixed(2)}</p>
63   )}
64
65   <div className="mt-auto flex gap-1">
66     <Link
67       to={` /product/${product.id}`}
68       className="flex-1 px-2 py-1.5 text-sm font-medium bg-slate-700 text-white text-center
69       ↪ rounded-md hover:bg-slate-800 transition-colors cursor-pointer"
70     >
71       View
72     </Link>
73     <button
74       onClick={() => onAddToCart(product)}
75       className="flex-1 px-2 py-1.5 text-sm font-medium bg-blue-600 text-white rounded-md
76       ↪ hover:bg-blue-700 transition-colors cursor-pointer"
77     >
78       Add to Cart
79     </button>
80   </div>
81 </div>
82 </article>
83 );
84 });

```

```

1 // generate_2000.js
2 const fs = require('fs');
3 const path = require('path');
4 let sharp = null;
5
6 try {
7   sharp = require('sharp');
8 } catch {
9   sharp = null;
10 }
11
12 const dataPath = path.join(__dirname, 'src', 'data.json');
13 const productsDir = path.join(__dirname, 'public', 'images', 'products');
14
15 function generateMassiveDataset() {
16   if (!fs.existsSync(dataPath)) {
17     console.error('Error: Cannot find src/data.json!');
18     return;
19   }
20 }

```

```

21 const rawData = fs.readFileSync(dataPath);
22 const data = JSON.parse(rawData);
23 const originalProducts = data.products;
24
25 if (!originalProducts || originalProducts.length === 0) {
26   console.error('Error: No products found to use as templates.');
```

↩ mode.');

```

27   return;
28 }
29
30 const templates = originalProducts.slice(0, 20);
31 const categories = ["Men's Clothing", "Women's Clothing", "Jewelery", "Electronics"];
32 const massiveProducts = [];
33 const totalProducts = 2000;
34
35 console.log('Generating 2000 products. This will just take a second...');
```

↩ mode.');

```

36
37 for (let i = 1; i <= totalProducts; i++) {
38   const template = templates[(i - 1) % templates.length];
39   const cleanTitle = template.title.replace(/\d+$/, '').trim();
40
41   const isInFirstRange = i >= 1 && i <= 200;
42   const isInSecondRange = i >= 201 && i <= 2000;
43   const hasDiscount = isInSecondRange || (isInFirstRange && Math.random() < 0.5);
44   const discountPercentage = hasDiscount ? Math.floor(Math.random() * 80) + 1 : null;
45
46   const nextProduct = {
47     id: i,
48     title: `${cleanTitle} ${i}`,
49     description: template.description,
50     price: template.price,
51     category: categories[(i - 1) % categories.length],
52     image: `/images/products/product-${i}.jpg`,
53     comments: []
54   };
55
56   if (discountPercentage) {
57     nextProduct.discountPercentage = discountPercentage;
58   }
59
60   massiveProducts.push(nextProduct);
61 }
62
63 data.products = massiveProducts;
64 fs.writeFileSync(dataPath, JSON.stringify(data, null, 2));
65 console.log('Done. data.json now has 2000 products.');
```

↩ mode.');

```

66 }
67
68 async function generateThumbnails() {
69   if (!sharp) {
70     console.error('Error: sharp is not installed. Install it first, then run this script with thumbs
71     ↩ mode.');
```

↩ mode.');

```

72     process.exit(1);
73   }
74
75   if (!fs.existsSync(productsDir)) {
76     console.error('Error: products image folder not found.');
```

↩ mode.');

```

77     process.exit(1);
78   }
79
80   const entries = fs.readdirSync(productsDir);
81   const sourceFiles = entries.filter((name) => {
82     const lower = name.toLowerCase();
83     return (
84       (lower.endsWith('.jpg') || lower.endsWith('.jpeg') || lower.endsWith('.png')) &&
85       !lower.includes('-thumb.'))
86     );
87   });
88
89   for (const fileName of sourceFiles) {
90     const inputPath = path.join(productsDir, fileName);
91     const baseName = fileName.replace(/\.([~/.])+$/, '');
92
93     const thumbJpg = path.join(productsDir, `${baseName}-thumb.jpg`);
```

```

93 const thumbWebp = path.join(productsDir, `${baseName}-thumb.webp`);
94 const thumbAvif = path.join(productsDir, `${baseName}-thumb.avif`);
95
96 const pipeline = sharp(inputPath).resize(350, 210, {
97   fit: 'contain',
98   background: { r: 255, g: 255, b: 255, alpha: 1 }
99 });
100
101 await pipeline.clone().jpeg({ quality: 82 }).toFile(thumbJpg);
102 await pipeline.clone().webp({ quality: 80 }).toFile(thumbWebp);
103 await pipeline.clone().avif({ quality: 50 }).toFile(thumbAvif);
104 }
105
106 console.log(`Thumbnails generated for ${sourceFiles.length} source images.`);
107 }
108
109 const mode = process.argv[2];
110
111 if (mode === 'thumbs') {
112   generateThumbnails().catch((error) => {
113     console.error(error);
114     process.exit(1);
115   });
116 } else {
117   generateMassiveDataset();
118 }

```

Engineering Critique 01: The sharp package was not installed. I got this error as well: React Hook useEffect has a missing dependency: 'debouncedSearchQuery'. Either include it or remove the dependency array in the Home.js

Prompt 20: "Add Sharp package and fix this error React Hook useEffect has a missing dependency: 'debouncedSearchQuery'. Either include it or remove the dependency array in the Home.js."

Copilot's Raw Output:

```

1 // package.json
2 "react-router-dom": "^7.13.1",
3 "react-scripts": "5.0.1",
4 "sharp": "^0.34.0",
5 "web-vitals": "^2.1.4"

```

```

1 // Home.js
2 }, [
3   appState.productsLoading,
4   appState.productsHasMore,
5   debouncedSearchQuery,
6   selectedCategory,
7   sortOption,
8   loadMoreProducts
9 ]);

```

Engineering Critique 01: Pictures had white background when resizing

Prompt 21: "I want the generated picture should be cropped and fixed by width and not height so it is ok some parts of the top and bottom of the picture is not resented but it is important that the picture covers entire 350*210 px and does not have any white background "

Copilot's Raw Output:

```

1 // generate_2000.js
2 const pipeline = sharp(inputPath).resize(350, 210, {
3   fit: 'cover',

```

```

4   position: 'centre'
5 });
6
7 await pipeline.clone().jpeg({ quality: 82 }).toFile(thumbJpg);
8 await pipeline.clone().webp({ quality: 80 }).toFile(thumbWebp);
9 await pipeline.clone().avif({ quality: 50 }).toFile(thumbAvif);

```

```

1 // Bash
2 node generate_2000.js thumbs

```

Engineering Critique 01: ok

Prompt 22: "How to resize this banner with script with width 1500px and use the thumbnail in the home page."

Copilot's Raw Output:

```

1 // Home.js
2 backgroundImage: "url('/images/banner-1500.jpg')"

```

```

1 // generate_2000.js
2 async function resizeBannerToWidth1500() {
3   if (!sharp) {
4     console.error('Error: sharp is not installed.');
```

```

5     process.exit(1);
6   }
7
8   const inputPath = path.join(__dirname, 'public', 'images', 'banner.jpg');
9   const outputPath = path.join(__dirname, 'public', 'images', 'banner-1500.jpg');
```

```

10
11   if (!fs.existsSync(inputPath)) {
12     console.error('Error: banner image not found at public/images/banner.jpg');
13     process.exit(1);
14   }
15
16   await sharp(inputPath)
17     .resize({
18       width: 1500,
19       withoutEnlargement: true
20     })
21     .jpeg({ quality: 85 })
22     .toFile(outputPath);
23
24   console.log('Banner resized to width 1500:', outputPath);
25 }

```

```

1 // generate_2000.js
2 const mode = process.argv[2];
3
4 if (mode === 'thumbs') {
5   generateThumbnails().catch((error) => {
6     console.error(error);
7     process.exit(1);
8   });
9 } else if (mode === 'banner-1500') {
10  resizeBannerToWidth1500().catch((error) => {
11    console.error(error);
12    process.exit(1);
13  });
14 } else {
15  generateMassiveDataset();
16 }

```

```
1 // Home.js
2 backgroundImage: "url('/images/banner-1500.jpg')"
```

```
1 // Bash
2 node generate_2000.js banner-1500
```

Engineering Critique 01: ok

Prompt 23: "I got a 'Use efficient cache lifetimes' red flag, so fix it
Configure static asset caching headers in server.js:
For fingerprinted static assets (js, css, media, images): Cache-Control should be public, max-age=31536000, immutable.
For index.html: Cache-Control should be no-cache (or max-age=0, must-revalidate).
Keep all existing API endpoints and behavior unchanged.
Ensure this works in production mode and does not break local development.
If needed, add separate behavior for static file caching vs API responses.
Do not introduce breaking changes to current routes."

Copilot's Raw Output:

```
1 // server.js
2 const SUPPORT_EMAIL = 'bakhshielias@gmail.com';
3 const publicDirPath = path.join(__dirname, 'public');
4 const buildDirPath = path.join(__dirname, 'build');
```

```
1 // server.js
2 const mimeByExtension = {
3   '.html': 'text/html; charset=utf-8',
4   '.js': 'text/javascript; charset=utf-8',
5   '.mjs': 'text/javascript; charset=utf-8',
6   '.css': 'text/css; charset=utf-8',
7   '.json': 'application/json; charset=utf-8',
8   '.map': 'application/json; charset=utf-8',
9   '.png': 'image/png',
10  '.jpg': 'image/jpeg',
11  '.jpeg': 'image/jpeg',
12  '.webp': 'image/webp',
13  '.avif': 'image/avif',
14  '.svg': 'image/svg+xml',
15  '.ico': 'image/x-icon',
16  '.txt': 'text/plain; charset=utf-8',
17  '.woff': 'font/woff',
18  '.woff2': 'font/woff2'
19 };
20
21 const getMimeType = (filePath) => {
22   const ext = path.extname(filePath).toLowerCase();
23   return mimeByExtension[ext] || 'application/octet-stream';
24 };
25
26 const getStaticCacheControl = (filePath) => {
27   const normalized = filePath.replace(/\\/g, '/');
28   const fileName = path.basename(normalized);
29
30   if (normalized.endsWith('/index.html') || normalized.endsWith('.html')) {
31     return 'no-cache, max-age=0, must-revalidate';
32   }
33
34   const hasFingerprint = /[.-][a-f0-9]{8,}\.\/i.test(fileName) || normalized.includes('/static/');
35   const isStaticAsset =
36     normalized.includes('/images/') ||
37     /\. (js|mjs|css|map|png|jpe?g|webp|avif|svg|ico|woff2?)$/i.test(normalized);
38
```

```

39   if (hasFingerprint || isStaticAsset) {
40     return 'public, max-age=31536000, immutable';
41   }
42
43   return 'public, max-age=3600';
44 };
45
46 const tryServeStaticFile = async (rootDir, pathname, res, method) => {
47   const cleanPathname = (() => {
48     try {
49       return decodeURIComponent(pathname || '/');
50     } catch {
51       return pathname || '/';
52     }
53   })();
54
55   const normalizedPath = cleanPathname === '/' ? '/' : cleanPathname.replace(/\/+$/, '');
56   const relativePath = normalizedPath === '/' ? '' : normalizedPath.replace(/~\/+/, '');
57
58   const candidates = [];
59   if (normalizedPath === '/') {
60     candidates.push('index.html');
61   } else {
62     candidates.push(relativePath);
63     if (!path.extname(relativePath)) {
64       candidates.push(path.join(relativePath, 'index.html'));
65     }
66   }
67
68   for (const candidate of candidates) {
69     const absolutePath = path.resolve(rootDir, candidate);
70     const safeRoot = path.resolve(rootDir) + path.sep;
71
72     if (absolutePath !== path.resolve(rootDir) && !absolutePath.startsWith(safeRoot)) {
73       continue;
74     }
75
76     try {
77       const stat = await fs.stat(absolutePath);
78       if (!stat.isFile()) {
79         continue;
80       }
81
82       const body = method === 'HEAD' ? null : await fs.readFile(absolutePath);
83       res.writeHead(200, {
84         'Content-Type': getMimeType(absolutePath),
85         'Cache-Control': getStaticCacheControl(absolutePath)
86       });
87
88       if (method !== 'HEAD') {
89         res.end(body);
90       } else {
91         res.end();
92       }
93
94       return true;
95     } catch {
96       // Try next candidate
97     }
98   }
99
100  if (!path.extname(relativePath)) {
101    const spaIndexPath = path.join(rootDir, 'index.html');
102    try {
103      const indexStat = await fs.stat(spaIndexPath);
104      if (indexStat.isFile()) {
105        const body = method === 'HEAD' ? null : await fs.readFile(spaIndexPath);
106        res.writeHead(200, {
107          'Content-Type': 'text/html; charset=utf-8',
108          'Cache-Control': 'no-cache, max-age=0, must-revalidate'
109        });
110
111        if (method !== 'HEAD') {

```



```

112     res.end(body);
113   } else {
114     res.end();
115   }
116
117   return true;
118 }
119 } catch {
120   // No SPA index fallback
121 }
122 }
123
124 return false;
125 };
126
127 const tryServeStaticRequest = async (req, res, pathname) => {
128   if (!['GET', 'HEAD'].includes(req.method)) {
129     return false;
130   }
131
132   if (pathname.startsWith('/api/')) {
133     return false;
134   }
135
136   if (await tryServeStaticFile(buildDirPath, pathname, res, req.method)) {
137     return true;
138   }
139
140   if (await tryServeStaticFile(publicDirPath, pathname, res, req.method)) {
141     return true;
142   }
143
144   return false;
145 };

```

```

1 // server.js
2 respond(res, 404, { message: 'Not found' });
3 });

```

```

1 // server.js
2 const wasStaticServed = await tryServeStaticRequest(req, res, pathname);
3 if (wasStaticServed) {
4   return;
5 }
6
7 respond(res, 404, { message: 'Not found' });
8 });

```

Engineering Critique 01: ok

Prompt 24: "Lighthouse is flagging LCP because your hero is a CSS background on a div so change it to normal picture in a div."

Copilot's Raw Output:

```

1 // Home.js
2 <div className="h-[320px] relative border-b border-slate-300 overflow-hidden">
3   
14   <div className="absolute inset-0 bg-slate-950/60" />
15   <div className="relative z-10 h-full flex flex-col justify-center px-6 md:px-10 text-white">
16     <h1 className="text-4xl md:text-5xl font-semibold mb-3 drop-shadow">
17       New Season Arrivals
18     </h1>
19     <p className="text-base md:text-lg text-slate-100 max-w-2xl">
20       Discover new products for your everyday shopping.
21     </p>
22   </div>
23 </div>

```

```

1 // index.html
2 <link
3   rel="preload"
4   as="image"
5   href="/images/banner-1500.jpg"
6   fetchpriority="high"
7 >/>

```

Engineering Critique 01: ok

Prompt 25: "got this red flag in the lighthouse 'elements contain focusable descendents' så remove all the aria-hidden."

Copilot's Raw Output:

```

1 // App.js
2 - <div className="mx-auto w-full max-w-[1500px] min-h-screen bg-white" aria-hidden="true">
3 + <div className="mx-auto w-full max-w-[1500px] min-h-screen bg-white">

1 // Home.js
2 -   aria-hidden="true"

```

Engineering Critique 01: ok

Prompt 26: "Lighthouse red flag : Document does not have a main landmark.
Add exactly one main landmark to the page structure.
Ensure primary page content is inside main and repeated layout elements (header/nav/footer) are outside main.
Keep existing routes and styles unchanged."

Copilot's Raw Output:

```

1 // App.js
2 <main>
3   <Routes>
4     <Route path="/" element={<Home />} />
5     <Route path="/sales" element={<Sales />} />
6     <Route path="/product/:productId" element={<ProductDetails />} />
7     <Route path="/contact" element={<Contact />} />
8     <Route path="/cart" element={<Cart />} />
9     <Route path="/checkout" element={<Checkout />} />
10    <Route
11      path="/login"
12      element={isLoggedIn ? <Navigate to="/profile" replace /> : <Login />}
13    />
14    <Route
15      path="/register"
16      element={isLoggedIn ? <Navigate to="/profile" replace /> : <Register />}

```

```

17     />
18     <Route
19       path="/profile"
20       element={
21         isLoggedIn ? (
22           <section className="p-8">
23             <div className="max-w-md mx-auto bg-white border border-slate-200 rounded-lg p-6
24               ↳ shadow-sm">
25               <h2 className="text-2xl font-semibold text-slate-800 mb-4">Profile</h2>
26               <p className="text-sm text-slate-800 font-semibold">Welcome {currentUser?.name ||
27                 ↳ appState.userAuthStatus.userEmail?.split('@')[0] || 'User'}</p>
28               <p className="text-slate-700 mb-4">{appState.userAuthStatus.userEmail}</p>
29
30               <div className="mb-4 border border-slate-200 p-2 max-h-48 overflow-y-scroll">
31                 <h3 className="text-base font-semibold mb-2">Order History</h3>
32                 {userOrders.length === 0 ? (
33                   <p className="text-xs text-slate-600">No order history found.</p>
34                 ) : (
35                   userOrders
36                     .slice()
37                     .reverse()
38                     .map((order) => (
39                       <div key={order.id} className="mb-2 border border-slate-300 p-2">
40                         <p className="text-xs">Order ID: {order.id}</p>
41                         <p className="text-xs">Placed: {order.placedAt}</p>
42                         <p className="text-xs">Total: ${order.total}</p>
43                         <p className="text-xs">Items: {Array.isArray(order.items) ?
44                           ↳ order.items.length : 0}</p>
45                       </div>
46                     ))
47                 )}
48               </div>
49             <button
50               className="w-full bg-slate-900 text-white py-2 rounded"
51               onClick={() => {
52                 logoutUser();
53                 navigate('/');
54               }}
55             >Logout</button>
56           </div>
57         </section>
58       ) : (
59         <Navigate to="/login" replace />
60       )
61     />
62   </Route>
63 </Routes>
</main>

```

Engineering Critique 01: ok

Prompt 27: "Lighthouse red flag : Document does not have a title element.

Add a valid title in index.html.

Keep title meaningful for the app and SEO-friendly.

Optionally improve route-level titles for key pages (Home, Product Details, Cart, Checkout) without adding heavy dependencies.

Do not break existing rendering or routing.

Return apply-ready code scopes and list final titles used."

Copilot's Raw Output:

```

1 // index.html
2 - <meta name="description" content="This is a React shop" />
3 + <meta
4 +   name="description"
5 +   content="My Ecommerce: shop fashion, electronics, and everyday essentials with fast, simple
↳ checkout."

```

```

6 + </>
7 + <title>My Ecommerce | Online Shopping for Fashion, Electronics and More</title>

```

```

1 // App.js
2 -import { useContext } from 'react';
3 -import { NavLink, Navigate, Route, Routes, useNavigate } from 'react-router-dom';
4 +import { useContext, useEffect } from 'react';
5 +import { NavLink, Navigate, Route, Routes, useLocation, useNavigate } from 'react-router-dom';

```

```

1 // App.js
2 + const location = useLocation();
3 +
4 + useEffect(() => {
5 +   const pathname = location.pathname;
6 +
7 +   if (pathname === '/') {
8 +     document.title = 'My Ecommerce | Home';
9 +     return;
10 +  }
11 +
12 +  if (pathname.startsWith('/product/')) {
13 +    document.title = 'My Ecommerce | Product Details';
14 +    return;
15 +  }
16 +
17 +  const titleByPath = {
18 +    '/cart': 'My Ecommerce | Cart',
19 +    '/checkout': 'My Ecommerce | Checkout',
20 +    '/sales': 'My Ecommerce | Sales',
21 +    '/contact': 'My Ecommerce | Contact',
22 +    '/login': 'My Ecommerce | Login',
23 +    '/register': 'My Ecommerce | Register',
24 +    '/profile': 'My Ecommerce | Profile'
25 +  };
26 +
27 +  document.title = titleByPath[pathname] || 'My Ecommerce | Online Store';
28 + }, [location.pathname]);

```

Engineering Critique 01: ok

Prompt 28: "I got this Lighthouse red flag 'Some elements have a [tabindex] value greater than 0' så remove all the tabindex attributes."

Copilot's Raw Output:

```

1 // App.js
2 <NavLink
3 -   tabIndex={7}
4     to="/"
5   end
6   className={({ isActive }) =>
7     `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
8     ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
9     ↪ hover:bg-slate-100'}`
10  }
11  >
12    Home
13  </NavLink>
14  <NavLink
15 -   tabIndex={8}
16     to="/sales"
17   className={({ isActive }) =>
18     `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
19     ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950

```

```

20     }
21   >
22     Sales
23   </NavLink>
24
25   <NavLink
26     tabIndex={2}
27     to="/cart"
28     className={({ isActive }) =>
29       `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
30       ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
31         ↪ hover:bg-slate-100'}`
32   }
33   >
34     Cart
35   </NavLink>
36
37   <NavLink
38     tabIndex={10}
39     to="/contact"
40     className={({ isActive }) =>
41       `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
42       ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
43         ↪ hover:bg-slate-100'}`
44   }
45   >
46     Contact
47   </NavLink>
48
49   {isLoggedIn ? (
50     <NavLink
51       tabIndex={11}
52       to="/profile"
53       className={({ isActive }) =>
54         `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200
55         ↪ cursor-pointer
56         ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
57           ↪ hover:bg-slate-100'}`
58     }
59   ) : (
60     <>
61       <NavLink
62         tabIndex={1}
63         to="/login"
64         className={({ isActive }) =>
65           `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200
66           ↪ cursor-pointer
67           ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800
68             ↪ hover:text-slate-950 hover:bg-slate-100'}`
69         }
70       >
71         Login
72       </NavLink>
73
74       <NavLink
75         tabIndex={9}
76         to="/register"
77         className={({ isActive }) =>
78           `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200
79           ↪ cursor-pointer
80           ${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800
81             ↪ hover:text-slate-950 hover:bg-slate-100'}`
82         }
83       >
84         Register
85       </NavLink>
86     </>
87   )}

```

Prompt 29: "remove all the tabIndex in the code "

Copilot's Raw Output:

```
1 // App.js
2 *** Begin Patch
3 *** Update File: src/App.js
4 @@
5 -     <NavLink
6 -         tabIndex={7}
7 -         to="/"
8 +     <NavLink
9 +         to="/"
10         end
11         className={({ isActive }) =>
12 @@
13 -     <NavLink
14 -         tabIndex={8}
15 -         to="/sales"
16 +     <NavLink
17 +         to="/sales"
18         className={({ isActive }) =>
19 @@
20 -     <NavLink
21 -         tabIndex={2}
22 -         to="/cart"
23 +     <NavLink
24 +         to="/cart"
25         className={({ isActive }) =>
26 @@
27 -     <NavLink
28 -         tabIndex={10}
29 -         to="/contact"
30 +     <NavLink
31 +         to="/contact"
32         className={({ isActive }) =>
33 @@
34 -     <NavLink
35 -         tabIndex={11}
36 -         to="/profile"
37 +     <NavLink
38 +         to="/profile"
39         className={({ isActive }) =>
40 @@
41 -     <NavLink
42 -         tabIndex={1}
43 -         to="/login"
44 +     <NavLink
45 +         to="/login"
46         className={({ isActive }) =>
47 @@
48 -     <NavLink
49 -         tabIndex={9}
50 -         to="/register"
51 +     <NavLink
52 +         to="/register"
53         className={({ isActive }) =>
54 *** End Patch
```

Engineering Critique 01: ok

Prompt 30: "I got main landmark. So add the main tag in the structure"

Copilot's Raw Output:

```
1 // App.js
2 <main>
3   <Routes>
4     <Route path="/" element={<Home />} />
```

```

5     <Route path="/sales" element={<Sales />} />
6     <Route path="/product/:productId" element={<ProductDetails />} />
7     <Route path="/contact" element={<Contact />} />
8     <Route path="/cart" element={<Cart />} />
9     <Route path="/checkout" element={<Checkout />} />
10    <Route
11      path="/login"
12      element={isLoggedIn ? <Navigate to="/profile" replace /> : <Login />}
13    />
14    <Route
15      path="/register"
16      element={isLoggedIn ? <Navigate to="/profile" replace /> : <Register />}
17    />
18    <Route
19      path="/profile"
20      element={
21        isLoggedIn ? (
22          <section className="p-8">
23            <div className="max-w-md mx-auto bg-white border border-slate-200 rounded-lg p-6
24              ↪ shadow-sm">
25              <h2 className="text-2xl font-semibold text-slate-800 mb-4">Profile</h2>
26              <p className="text-sm text-slate-800 font-semibold">Welcome {currentUser?.name ||
27                ↪ appState.userAuthStatus.userEmail?.split('@')[0] || 'User'}</p>
28              <p className="text-slate-700 mb-4">{appState.userAuthStatus.userEmail}</p>
29
30              <div className="mb-4 border border-slate-200 p-2 max-h-48 overflow-y-scroll">
31                <h3 className="text-base font-semibold mb-2">Order History</h3>
32                {userOrders.length === 0 ? (
33                  <p className="text-xs text-slate-600">No order history found.</p>
34                ) : (
35                  userOrders
36                    .slice()
37                    .reverse()
38                    .map((order) => (
39                      <div key={order.id} className="mb-2 border border-slate-300 p-2">
40                        <p className="text-xs">Order ID: {order.id}</p>
41                        <p className="text-xs">Placed: {order.placedAt}</p>
42                        <p className="text-xs">Total: ${order.total}</p>
43                        <p className="text-xs">Items: {Array.isArray(order.items) ?
44                          ↪ order.items.length : 0}</p>
45                      </div>
46                    ))
47                )}
48              </div>
49
50              <button
51                className="w-full bg-slate-900 text-white py-2 rounded"
52                onClick={() => {
53                  logoutUser();
54                  navigate('/');
55                }}
56              >Logout</button>
57            </div>
58          </section>
59        ) : (
60          <Navigate to="/login" replace />
61        )
62      )
63    />
64    <Route path="*" element={<Navigate to="/" replace />} />
65  </Routes>
66 </main>

```

Engineering Critique 01: ok

Prompt 31: "I click on the "apply in editor" but the code does not move and apply to the old code."

Copilot's Raw Output:

```
1 // AppContext.js
```

```

2 @@
3   const addToCart = (product) => {
4     setAppState((prev) => ({
5       ...prev,
6       shoppingCart: [...prev.shoppingCart, product]
7     }));
8     window.alert('Added to cart');
9   };

```

```

1 // App.js
2 @@
3   function AppContent() {
4     const { appState, logoutUser } = useContext(AppContext);
5     const navigate = useNavigate();
6     const location = useLocation();
7   @@
8     const isLoggedIn = appState.userAuthStatus.isAuthenticated;
9 +   const cartCount = appState.shoppingCart.length;
10    const currentUser = appState.users.find(
11      (user) => (user.email || '').toLowerCase() === (appState.userAuthStatus.userEmail ||
12        ↪ '').toLowerCase()
13    );
14    @@
15    <NavLink
16      to="/cart"
17      className={({ isActive }) =>
18        ↪ `px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200 cursor-pointer
19        ↪ `relative px-3 py-1.5 text-sm font-semibold rounded-md transition-colors duration-200
20        ↪ `cursor-pointer
21        ↪ `${isActive ? 'bg-slate-900 text-white shadow-sm' : 'text-slate-800 hover:text-slate-950
22        ↪ ` hover:bg-slate-100'}`
23      }
24    >
25      Cart
26      {cartCount > 0 ? (
27        ↪ <span className="absolute -top-2 -right-2 min-w-5 h-5 px-1 rounded-full bg-blue-600
28        ↪ text-white text-[11px] leading-5 text-center font-bold">
29        ↪ {cartCount > 99 ? '99+' : cartCount}
30        ↪ </span>
31        ↪ ) : null}
32    </NavLink>

```

```

1 // Home.js
2 @@
3   <img
4     src={imageSrc}
5     alt={product.title}
6     ↪ className="h-52 w-full object-contain p-4"
7 +   className="h-52 w-full object-cover"
8     width={350}
9     height={210}
10    loading="lazy"
11    decoding="async"
12    onError={handleImageError}
13  />

```

Engineering Critique 01: ok

*Prompt 32: "create distans between procudt picture and the title
remove the description p tag in the product item "*

Copilot's Raw Output:

```

1 // Home.js
2 @@
3   <div className="px-4 pb-4 flex flex-col grow">

```



```

4 -      <h3 className="text-lg font-semibold text-slate-900 truncate">{product.title}</h3>
5 -      <p className="text-base text-slate-700 line-clamp-2 min-h-[48px]">{product.description}</p>
6 +      <div className="px-4 pb-4 flex flex-col grow">
7 +      <h3 className="mt-4 text-lg font-semibold text-slate-900 truncate">{product.title}</h3>

```

Engineering Critique 01: ok

3. The Final Verdict

Lighthouse Metrics (CPU 4x Throttling, Network Slow 4G):

Performance: 91 (*Up from 43.6*)

Accessibility: 100 (*Up from 29*)

First Contentful Paint (FCP): 1.28 s (*Down from 2.01*)

Largest Contentful Paint (LCP): 1.28 s (*Down from 3.8*)

Total Blocking Time (TBT): 0 (*Down from 627*)

Cumulative Layout Shift: 0

Speed Index: 1.54 s (*Down from 2.76*)

4. Engineering Conclusion

Copilot successfully fixed these problems and improved the performance and accessibility metrics significantly.

Major Improvements Delivered

- Upgraded page container, background, and readability styling for a cleaner visual baseline.
- Improved metadata and document semantics (language, description, title behavior).
- Refined product card UI, button hierarchy, spacing, and contrast for usability.
- Moved filtering/sorting/pagination to server query layer to reduce client computation.
- Added scalable product paging contract: q, category, sort, page, limit with hasMore.
- Implemented infinite scroll with IntersectionObserver and duplicate-request protection.
- Added loading skeletons, retry path, and clearer loading/no-more states.
- Introduced debounced search to avoid request storms on each keystroke.
- Optimized image delivery with generated thumbnails (AVIF/WebP/JPG fallback), lazy loading, and stable dimensions.
- Added thumbnail generation and banner resize tooling using Sharp.
- Improved static caching policy for build assets and HTML revalidation behavior.
- Replaced hero CSS background with image element for better LCP and preload compatibility.
- Fixed accessibility blockers: removed invalid positive tabindex usage and added one main landmark.
- Removed add to cart alert.

Resolved Regressions During Iteration

- Undefined sortProducts reference after refactor.
- Duplicate product loading behavior and initial flash inconsistencies.
- 404 mismatch for paged product endpoint handling.
- Missing React hook dependency warning.
- Missing Sharp package for image pipeline scripts.
- Thumbnail resize behavior adjusted from contain to cover to remove white bars.